19. Why is LIFO appropriate for checking nested brackets?

LIFO works because the last opening bracket must be closed first. The stack keeps track of the most recent opening bracket and checks whether the closing bracket matches it.

20. Why must the right operand be popped before the left operand?

The right operand is popped first because it is on top of the stack. The left operand is popped next. This keeps the correct order of the arithmetic operation, especially for subtraction and division.

21. What is the difference between balanced brackets and a valid arithmetic expression?

Balanced brackets mean that all opening and closing brackets match correctly. A valid arithmetic expression must also follow the correct order of numbers and operators. An expression can have balanced brackets but still be invalid.

22. What is the difference between a Stack ADT and a function call stack? Does a nested call return first?

A Stack ADT is a data structure that follows the Last In, First Out principle. A function call stack is used by a program to manage function calls and their execution. Yes, a nested function call returns first before the function that called it continues.

23. How is memory allocated and released in your implementation?

The program initially allocates memory for two elements using a dynamic array. When the stack becomes full, it creates a larger array with double the capacity and copies the existing elements. The old array is deleted to free memory. The destructor also releases the allocated array when the stack is destroyed.

Reflection

Through this activity, I learned how a stack can be used in different expression-processing tasks. I also understood how dynamic arrays manage memory as the stack grows. Testing valid and invalid expressions helped me understand the importance of checking input before performing calculations.